# 605.486 – Android Development

## Module 10a

## Custom Views and Basic Drawing

# Custom View (1)

- Create a class that extends android.view.View

- Must implement three constructors
  - Called from your code or when XML is inflated
  - Call super for all three
  - Custom XML attributes if desired

# Custom View (2)

- Override onDraw()
  - Performs drawing whenever refresh needed
  - Order of painting determines z-order
    - "Painter's algorithm" – last painted covers

- width/height not available in constructors
  - Graphical environment not yet ready
  - Sizes not yet set

# Using Custom View

- In layout xml, use one of
  - <view class="a.b.c.CustomView" … />
  - <a.b.c.CustomView …/>

# onDraw() caution!

- onDraw is called VERY often

- Should try to never create objects inside
  - Big mistake is creating instances of Paint
  - Allocate them in advance
  - Get resources during initialization/construction
  - If require sizing, use onSizeChanged()

# Custom View Attributes

- Define attributes in attrs.xml
- Fetch attributes in your View's constructor
- Add attributes to XML layout definition

# Animation

- Can have onDraw() call invalidate()
  - Every draw schedules a redraw…
  - Not a good idea…
- Don't sleep in onDraw!
  - Makes UI much less responsive to user

# Using a Separate Thread

- Start a separate thread to update positions of objects to animate
- Call postInvalidate() from that thread
  - puts request to invalidate() on the UI queue
- Result: More responsive UI

# Using Paths and Strokes

- Can define arbitrary Path objects
  - moveTo(x,y)
  - lineTo(x,y)
  - close()

- strokeWidth
- style (STROKE / FILL)
- strokeJoin (ROUND / BEVEL/ MITRE)

# Setting Custom View Sizes

- How to deal with explicit sizes or wrap_content / match_parent

- Override onMeasure()
  - DO NOT CALL super.onMeasure()!!!
  - Determine sizing
  - Call setMeasuredDimension(width, height)

# Determining View Sizes

- onMeasure passed two “measureSpecs”
  - Two parts – drawingMode and value
    - AT_MOST – parent container has hard limit
    - EXACTLY – size was explicitly set
    - UNSPECIFIED – no restrictions
  - Parse using MeasureSpec class to separate out values